6.5.3 Using ultrasound

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	ultrasound; longitudinal wave with frequency greater than 20 kHz	
(b)	piezoelectric effect; ultrasound transducer as a device that emits and receives ultrasound	
(c)	ultrasound A-scan and B-scan	HSW9, 10, 12

(e) reflection of ultrasound at a boundary; *M0.3*

$$\frac{I_r}{I_0} = \frac{(Z_2 - Z_1)^2}{(Z_2 + Z_1)^2}$$

(f) impedance (acoustic) matching; special gel used in ultrasound scanning

(g) Doppler effect in ultrasound; speed of blood in the patient; $\frac{\Delta f}{f} = \frac{2v\cos\theta}{c}$ for determining the speed v of blood.